NORTEX Code

User Manual

Version 0.1.0

Table of contents

1 Welcome to Nortex Code 4
- What is Nortex Code? 4
- What is it used for? 4
- Overview of the main features 4
- Relation to the language 4

2 Installation and first start 5
- System requirements 5
- Installation 5
- Starting 5
- First start 5
- Files written by the IDE 5
- Common problems 5

3 User interface 6
- Menu bar 6
- Activity bar 6
- Explorer 7
- Editor area 7
- Tabs 7
- Console (bottom panels) 7
- Sidecar 7
- Status bar 7
- Other views 7

4 Projects and files 8
- Creating a project 8
- Opening / refreshing a project 8
- Files 8
- Not available 8

5 Working with the editor 9
- Basic features 9
- Auto-completion 9
- Line commands 9
- Find and replace 9
- Tabs and split editor 10
- Navigation 10
- Rename, breadcrumbs and minimap 10

6 Build and run 11
- Triggering 11
- Sequence 11
- Reading the result 11
- Overflow checks 11
- Limits 11

7 Terminal and consoles 12
- Terminal 12

8 Error analysis and debugging 13

Recognizing errors 13
Reading diagnostics 13
Common problems 13
Debugging features 13
9 Settings and customization 15
Layout of the settings view 15
Reset, export and import 15
Active settings 15
Planned (visible, but greyed out and without effect) 16
Shortcuts 16
Account and synchronization 16
Window layout 16
Not customizable 17
10 Keyboard shortcuts 18
General 18
File management 18
Editor 18
Navigation and side bar 19
Terminal and console 19
Build, run and debugging 19
Panels and windows 19
Menu shortcuts 19
11 First steps 20
12 Efficient workflows 21
13 FAQ and troubleshooting 22

1 Welcome to Nortex Code

Contents · Next: Installation

What is Nortex Code?

Nortex Code is the graphical development environment (IDE, "Integrated Development Environment") for the Nortex programming language. An IDE combines a text editor, file overview, compiler invocation and output in a single window.

Nortex Code is a standalone Windows program. To build, it calls the Nortex compiler nortexc, which translates .nx files into a Windows executable (EXE).

What is it used for?

- Writing Nortex source code (.nx),
- building and running with one click (Run, Ctrl+F5),
- reading compiler messages with their positions,
- editing several files in tabs and split views.

Overview of the main features

Area	Feature
Editor	Syntax highlighting, line numbers, auto-completion, find/replace, split view
Build	Run button, overflow checks, nortex.toml projects
Output	Compiler Log, Runtime Output, Terminal (PowerShell)
Customization	Theme, density, UI and editor zoom, language (German/English)

Not included are, among others, a full debugger backend, project wizards and file rename/delete (see feature status).

Relation to the language

The language determines what can be compiled. Today the compiler natively builds only a small subset (including int/bool, if/while/for, functions and UTF-8 string slices for variables, function calls and Console.writeLine). The IDE recognizes more syntax than the build can actually produce; details: Language reference.

2 Installation and first start

Back · Contents · Next: Interface

System requirements

- Windows 10/11, 64-bit.
- To link generated programs: the Microsoft linker (Visual Studio Build Tools) – see Language reference, chapter 2.
- Optional: Git (for the Git view) and PowerShell (for the terminal).

Installation

Unpack the program folder anywhere. Next to nortex-code.exe there must be the compiler nortexc.exe and the library nortex_runtime.lib. If something is missing, Build & Run reports it in the Compiler Log; the dashboard shows the state of the tools.

Starting

Double-click nortex-code.exe. The working directory at start becomes the project folder (shown in the Explorer field). If you start the EXE from a shortcut, set the desired "Start in" folder there.

First start

On first start the dashboard opens (project overview with the buttons New, Open file, Build & Run and "Toolchain Diagnostics"). It shows whether the compiler and the runtime library were found. Both should be green.

[Abbildung: Dashboard]

Files written by the IDE

- Your settings (theme, density, zoom, language, recently opened files) in the folder %APPDATA%\NORTEX Code.
- When building: .exe next to the source file (without a manifest) or the output location given in nortex.toml.

Common problems

Problem	Solution
Compiler not found	Put nortexc.exe next to nortex-code.exe
Runtime library not found	Put nortex_runtime.lib next to the EXE
Linker error during build	Install the Visual Studio Build Tools
Terminal does not start	Message "Shell cannot be started"; PowerShell must be available

3 User interface

Back · Contents · Next: Projects and files

[Abbildung: Main window with editor]

No.	Area	Location
1	Menu bar	top
2	Activity bar	far left (narrow icon strip)
3	Side bar (Explorer, Search, Git, Debugger)	left of the activity bar
4	Editor area with tabs and run bar	center
5	Sidecar	right
6	Status bar	between editor and console
7	Console (Compiler Log, Runtime Output, Terminal)	bottom

Menu bar

Menus: File, Edit, View, AI, Help.

Menu	Entries
File	New file (Ctrl+N) · Open… · Open folder… · Save (Ctrl+S) · Save as… · Close project folder (only hides the side bar) · Refresh project · Open project folder in Explorer · Exit
Edit	Undo · Redo (greyed out when there is nothing to do) · Cut (Ctrl+X) · Copy (Ctrl+C) · Paste (greyed out, only via Ctrl+V in the editor) · Select all (Ctrl+A) · Find (Ctrl+F) · Find/Replace (Ctrl+H) · Duplicate line · Move line up/down · Select next occurrence · Toggle block comment (Ctrl+/) · Go to line…
View	Zen mode (F11) · Explorer (Ctrl+B) · Terminal / Diagnostics (Ctrl+J) · Show status bar · Side bar (Explorer, Search, Git, Debugger) · Reset layout · Show line numbers · Show minimap (Ctrl+Alt+M) · Reset zoom · Sidecar (Ctrl+Shift+L) · Editor layout (Single, Two columns, Two rows, 2x2 grid, Close pane) · Theme · Density · UI zoom · Editor zoom · Language
AI	Note "Not connected yet – no code is sent."; all entries (Explain code, Refactoring suggestion, Analyze error, Suggest tests, Generate documentation) are disabled, planned
Help	Getting started · Copy diagnostic information · License · NORTEX Code manual · Open documentation folder · Keyboard shortcuts · Update check (only reports that the check is unavailable offline) · About

[Abbildung: Edit menu]

Activity bar

At the top the N logo (dashboard), below it Explorer, Editor, Search, Git, Debugger, and at the very bottom the gear (settings). The active icon has a 2 px strip on its left edge (purple for Explorer/Editor/Dashboard, blue for Git/Debugger). Clicking the already active side icon toggles the side bar (Ctrl+B).

Explorer

Shows the project folder as a tree with collapsible folders. The icon color depends on the file type (.nx purple). A click opens the file. At the top: an input field for the project folder and Open.

Editor area

Top row: Run (Ctrl+F5), Save, the checkbox Overflow checks, file name. Tabs and the editor follow. File -> Save as… opens the file dialog. Details: Chapter 5.

Tabs

One tab per open file; + creates a new document. The active tab has a lighter background and an amber line. Unsaved changes are shown by an amber dot • next to the always visible x (close). With unsaved changes a dialog asks whether to save or discard. A red dot ? marks the active tab when there are compiler errors. Tabs can be dragged by their file name to reorder them or to move them to another pane.

Console (bottom panels)

Tabs Compiler Log (n), Runtime Output, Terminal. Toggle with Ctrl+J or View -> Terminal / Diagnostics. See Chapter 7.

Sidecar

Right panel "AI assistance" with an input field and Send. It is not connected to any AI provider. Toggle with Ctrl+Shift+L or View -> Sidecar. In narrow windows it collapses automatically.

Status bar

From left to right: project name · status message (green after a successful run, red on errors) · error count · line and column. It can be hidden under View -> Status bar.

Other views

- Dashboard: project overview, build and toolchain status, recently opened files.
- Settings: gear icon (Chapter 9).
- Command palette: Ctrl+Shift+P or F1 (Chapter 10).

[Abbildung: Command palette]

Zen mode (View menu, F11) hides secondary areas; Esc exits it. There are no context menus (right-click). Small windows exist for Go to line, Help -> License, Keyboard shortcuts, About, the manual and the account window; the Windows file dialog is used for opening/saving files.

4 Projects and files

Back · Contents · Next: Editor

A project is a folder. There is no "New project" wizard.

Creating a project

1. Create a folder in Windows Explorer.
2. Optionally place a nortex.toml in it (manifest; see the manifest schema).
3. Start Nortex Code with this folder as working directory, or enter the path in the field at the top of the Explorer and click Open.

Opening / refreshing a project

Enter the path in the Explorer field -> Open. Status bar message "Project loaded" or "Project could not be refreshed: …". File -> Refresh project re-reads the file list in the background (also after changes made outside the IDE). With a nortex.toml in the project folder the status bar shows "Name -> output file" and the window title reads " — NORTEX Code".

Files

Action	How
New	+ in the tab area, File -> New file or New on the dashboard
Open	Click in the Explorer · Ctrl+P (Windows file dialog, filter .nx/.nortex) · Open file on the dashboard · drag a file onto the window · recently opened file on the dashboard
Save	Ctrl+S or Save. For a new document the save dialog opens.
Save as	File -> Save as… or Ctrl+Shift+S opens the file dialog.
Close	x on the tab. With unsaved changes you can save, discard or cancel.

Saving is atomic (a failed write leaves the original file unchanged). Errors appear as a status message, e.g. "Save failed: …".

Not available

Renaming, deleting, creating folders and moving files only work outside the IDE (Windows Explorer, Terminal tab). Afterwards use Refresh project.

5 Working with the editor

Back · Contents · Next: Build and run

Basic features

- Syntax highlighting for keywords, identifiers, numbers, strings, characters, operators, punctuation and comments (colors depend on the theme).
- Line numbers in the left column.
- Undo/Redo: Ctrl+Z, Ctrl+Y or Ctrl+Shift+Z (also in the Edit menu).
- Errors are marked directly in the editor (inline diagnostics, updated shortly after typing) and listed in the Compiler Log (Chapter 8).
- There is no bracket completion and no multi-cursor.

Auto-completion

From 2 characters of an identifier a popup opens with up to 5 suggestions: keywords first, then identifiers from the current document.

Key | Effect

Arrow up/down | Move selection (wraps around)

Tab or Enter | Accept suggestion

Esc | Close popup

Line commands

Command | Key

Toggle comment | Ctrl+/

Move line(s) up/down | Alt+^ / Alt+v

Duplicate line(s) | Shift+Alt+v

Select next occurrence of the selected text | Ctrl+D

Find and replace

Ctrl+F opens a floating field at the top right of the editor, Ctrl+H additionally with "Replace with". If text is selected it is taken over.

Control | Function

Search field | Matches are searched live

Switches Aa / W / .* | Match case · Whole word · Regular expressions (with capture groups in the replacement text)

? / ? or Enter / Shift+Enter | previous / next match

Replace / All | replace current / all matches

x or Esc | close

An invalid regular expression is shown as an error without crashing the IDE. Search across files: Search side bar (Ctrl+Shift+F): enter a term, Search or Enter; clicking a match opens the file (simple text search, no line highlighting).

Tabs and split editor

Several files live in tabs; Ctrl+Tab switches to the next tab. Via View -> Editor layout (or keyboard) the editor can be split:

Layout	Key
Two columns	Ctrl+\ (also Ctrl+Shift+2)
Two rows	Ctrl+Shift+\ (also Ctrl+Shift+3)
2x2 grid	Ctrl+Shift+4
Single	View -> Editor layout -> Single

Focus: Ctrl+1 … Ctrl+4 selects the pane; Ctrl+Alt+Arrow or Alt+<-/-> moves in that direction; Ctrl+Shift+W closes the active pane.

Navigation

There is no "Go to definition". Use: Ctrl+F, project search, tabs, Ctrl+P (open file), and Edit -> Go to line….

Rename, breadcrumbs and minimap

- F2 opens an input field at the cursor with the identifier under the cursor. Enter renames all occurrences in the current file, Esc cancels. Comments, strings and keywords are untouched; ASCII names only, no project-wide rename.
- Breadcrumbs above the editor show the file and enclosing scopes (e.g. struct and function). A click jumps to the start line of the scope.
- Minimap at the right edge (60 px); a click scrolls. Toggle with Ctrl+Alt+M, via the View menu or the command palette; stored as editor.minimap.enabled.

6 Build and run

Back · Contents · Next: Terminal and consoles

Nortex Code has one build function: Build & Run. It saves, compiles with nortexc build and starts the generated EXE.

Triggering

Way	Note
? Run button	in the editor area
Ctrl+F5	
Ctrl+Shift+B	same function (no build-only mode)
Command palette -> Build & Run	
Dashboard -> Build & Run	

While building the bar shows "Compiling…"; a second build cannot start at the same time.

Sequence

1. The active file is saved. For new documents without a path the save dialog appears.
2. Without nortex.toml: nortexc build --output --overflow-checks=; the EXE is placed next to the source file.
3. With nortex.toml in the project folder: the manifest is built. The EXE is placed in the project's build\ subfolder; overflow checks come from the IDE setting unless the manifest defines them.
4. The compiler log shows the compiler output. Then the IDE starts the EXE; its output appears in Runtime Output, followed by "Program finished (exit code …)".

[Abbildung: Result of a run]

Reading the result

- Status bar green: build succeeded. Red: compiler errors. The program's exit code is just a value (Nortex programs return it via return) and never turns red. In the screenshot "Exitcode Some(0)" is the IDE's actual display for exit code 0.
- On errors: Compiler Log tab (Chapter 8).

Overflow checks

The checkbox controls --overflow-checks. On: integer overflow causes a trap (the program aborts). Off: overflow wraps around. Default: on in debug builds, off in release builds of the IDE. The same option is in the settings and on the dashboard.

Limits

- Running programs cannot be cancelled (Shift+F5 only shows a notice).
- No input channel for running programs.
- Only the natively supported language subset builds (Language reference); everything else ends with NX2001/NX2002.

7 Terminal and consoles

Back · Contents · Next: Error analysis and debugging

The bottom panel 03 / CONSOLE has three tabs:

Tab	Content	Open directly
Compiler Log (n)	Compiler output and diagnostics; n = count	Ctrl+Shift+M
Runtime Output	Output of the started program and exit code	Ctrl+Shift+U
Terminal	real PowerShell session	click the tab

Show/hide the panel: Ctrl+J or View -> Terminal / Diagnostics. A new build clears Compiler Log and Runtime Output. Command palette -> Clear Console clears both as well.

Terminal

[Abbildung: Terminal]

- When the tab is first opened the shell (powershell.exe) starts in the background; the line shows "(active)" or "(ended)" with a green/red dot.
- Type a command in the lower input field, Enter sends it. The output appears above.
- Clear or Ctrl+L clears the display.
- It is not a full console: no interactive programs, no color codes.
- Only one terminal; multiple consoles or terminal tabs do not exist.
- If the shell does not start, "Shell cannot be started: …" is shown.

8 Error analysis and debugging

Back · Contents · Next: Settings

Recognizing errors

- The editor checks the text shortly after typing and marks errors directly in the code.
- The Compiler Log (n) tab lists all diagnostics; the status bar shows "n errors".
- A red dot ? in the active tab means: the file has errors.
- After a build the status message is red/green.

Reading diagnostics

Format in the Compiler Log: Code [file:line:column] message, below it optionally Help: …. Red = error, amber = warning.

Example (typo summee, line 3, column 12):

```
NX1004 [main.nx:3:12] Unknown identifier 'summee'. The value was neither declared nor impo
rted.
  Help: Check the spelling or declare the value before using it.
```

Fix: correct the name (summe). A missing semicolon is reported as NX0100 ("A ';' is expected after a variable declaration."); the position points to the next token, so the semicolon is usually missing on the line before.

Meaning of codes: Diagnostic codes and Language reference, chapter 11. Compiler messages themselves are currently in German.

Common problems

Message | Cause / solution

NX2001 / NX2002 | Construct is checked but not generated natively yet (e.g. float, string variables, structs). Restrict yourself to the supported subset

Linker error LNK2019 | public int main() is missing

"Compiler … could not be started" | nortexc.exe not found (Chapter 2)

"Build failed (exit code …)" | The errors are in the Compiler Log above

Debugging features

Real debugging (pausing, stepping, inspecting variables) is not available. Reason: nortexc does not emit debug information (e.g. line tables). Without it the IDE cannot stop a running program at a source line. The IDE says so openly instead of faking a session.

Breakpoint management and its user interface exist:

Feature | Status

Set/remove breakpoint | F9 on the cursor line. The file must be saved (new documents without a path: notice in the status bar). The status bar reports "Breakpoint set/removed at line n".

Marker in the editor | A ? in the column left of the line number marks lines with a breakpoint.

Breakpoint list | Debugger panel (Activity bar -> Debugger or Ctrl+Shift+D): "Breakpoints (n)" with file:line.

A click jumps to the line, x removes an entry, Remove all clears the list.

Call stack / variables | The panel contains the sections Call stack and Variables. They show "No active session." because no debugger is connected yet.

F5 / ? Debug | Opens the Debugger panel and reports that step-by-step debugging is unavailable. The session stays inactive.

? Stop | Ends the (non-existent) session state.

Shift+F5 | Reports "Cancelling running programs is not implemented."

Breakpoints are only held in the running IDE and are not saved; after a restart they are gone. They have no effect on Ctrl+F5 (Build & Run).

Troubleshooting currently works through diagnostics, program output and the exit code (e.g. return of an intermediate result, see Chapter 11).

9 Settings and customization

Back · Contents · Next: Keyboard shortcuts

[Abbildung: Settings]

Open: gear at the bottom of the activity bar (clicking again returns to the editor). The appearance controls are also in the View menu.

Layout of the settings view

- Search (top): searches title, description and keywords. All typed words must match. The search also filters the shortcut table.
- Level filter: All levels, User, Project or Advanced. User are personal defaults. Project is intended for project-specific values; these entries are prepared but without effect yet (see "Planned"). Advanced contains special options.
- 7 categories: Appearance · Editor · Shortcuts · Language and code help · Build and debug · Account and sync · System, security and updates.
- Changed entries carry a dot •; ? resets the single entry to its default.

Reset, export and import

Action | Effect

? on an entry | resets this entry

Reset category | resets all active settings of the category

Reset all | resets all active settings; the list of recently opened files is kept

Export… | saves the settings as JSON (suggested name nortex-code-settings.json)

Import… | reads a JSON file; invalid files are rejected with a message, invalid single values fall back to the default. Your own list of recently opened files stays unchanged

Settings are saved automatically in the folder %APPDATA%\NORTEX Code (the path is shown at the end of the view). Damaged files are repaired at start.

Active settings

Setting | Category | Default | Effect

Color scheme | Appearance | Industrial Midnight | Industrial Midnight, Professional Dark or High Contrast. Command palette -> Toggle Theme switches to the next one

Density | Appearance | Comfortable | Compact, Comfortable or Spacious: spacing of the interface

Language | Appearance | German | German or English; applies immediately and is saved

Interface zoom | Appearance | 100 % | 70–200 %; Ctrl+mouse wheel changes it in 10 % steps

Editor zoom | Appearance | 100 % | 70–200 %, editor font

Show status bar | Appearance | on | shows/hides the status bar

Show line numbers | Editor | on | line number column

Show minimap | Editor | off | also Ctrl+Alt+M

Auto save | Editor | off | saves modified documents with a file path at most every 30 seconds; new documents without a path are not saved

Overflow checks | Build and debug | on in debug builds of the IDE, off in release builds | Trap on integer overflow; a nortex.toml can override the value. Same option as in the editor area (Chapter 6)

The zoom slider works in 10 % steps and takes effect on release; enter more precise values directly. Typed percentages are only applied with Enter or after leaving the field. Invalid values in the file fall back to the default on load.

Planned (visible, but greyed out and without effect)

These entries are shown with the note "planned"; the tooltip names the reason.

- Editor: tab width, word wrap, show whitespace, format on save, highlight bracket pairs, editor font
- Language and code help: live diagnostics, auto-completion (both are always on and cannot be disabled yet), inline hints
- Build and debug: default build profile, additional compiler arguments, working directory (project level missing), process time limit, stop on start (debugger not available yet, see Chapter 8)
- Account: synchronize settings
- System: update channel, usage data, workspace trust. Regarding usage data: no data is currently collected.

Shortcuts

The category shows the current assignment as a table. Customizing is not possible yet. Full list: Chapter 10.

Account and synchronization

- Local mode: There is no account system in the sense of an online service. The IDE runs entirely locally; editor, compiler, build and projects need no account.
- The Account menu item and the button Open Nortex account… show the status "Local mode". Sign in with browser reports that the account service is not available yet. No sign-in is faked.
- Synchronization: Four switches (settings, themes, shortcuts, list of extensions) are only prepared. They are saved but only take effect after a sign-in, which does not exist yet. All are off by default.
- Source code stays local: source code, projects and files never leave the computer; they are not among the synchronizable data.
- Planned (only visible as greyed-out entries): AI features, extensions, license, devices. A later sign-in will use the system browser, without passwords in the IDE.

Window layout

- Side bar: Ctrl+B · Console: Ctrl+J · Sidecar: Ctrl+Shift+L · Zen mode: F11 or View.
- Editor layout: Chapter 5.
- Panel widths and heights can be adjusted within limits by dragging. Panels cannot be freely moved or docked.

Not customizable

Key bindings, font, colors of individual elements and panel position (side bar always left, sidecar right, console bottom).

10 Keyboard shortcuts

Back · Contents · Next: First steps

The keys apply while the command palette is closed. The assignment cannot be changed. Shortcuts marked * do not appear in the help overview.

General

Key	Function
Ctrl+Shift+P, F1	Command palette
Ctrl+Z	Undo
Ctrl+Y, Ctrl+Shift+Z*	Redo
Ctrl+Shift+X*	Extensions – only shows "Extensions: not implemented."
F11	Toggle Zen mode
Esc	Leave Zen mode (when active)

In the palette: typing filters, arrow keys select, Enter runs, Esc closes. Available commands: Build & Run, Toggle Theme, Clear Console, Toggle Zen Mode, Open File, Save File, Split Editor Columns / Rows / 2x2 Grid, Editor Single Layout, Close Editor Pane, Toggle Minimap.

File management

Key	Function
Ctrl+N	New file
Ctrl+S	Save
Ctrl+Shift+S	Save as (file dialog)
Ctrl+P	"Quick open": opens the Windows file dialog (no searchable file list)
Ctrl+Tab*	next tab

Editor

Key	Function
Ctrl+F	Find
Ctrl+H	Replace
Ctrl+/	Toggle line comment
Ctrl+D	select next occurrence
Alt+^ / Alt+v	Move line
Shift+Alt+v	Duplicate line
F2	Rename symbol (current file)
Ctrl+Alt+M	Toggle minimap
Ctrl+Alt+B	Insert main skeleton
^ v, Tab/Enter, Esc	in the suggestion popup: select, accept, close
Enter, Shift+Enter, Esc	in the search field: next, previous match, close

Navigation and side bar

Key | Function

Ctrl+Shift+E* | Explorer

Ctrl+Shift+F* | Search

Ctrl+Shift+G* | Git

Ctrl+Shift+D* | Debugger panel

Ctrl+1 … Ctrl+4* | Editor pane 1–4

Ctrl+Alt+<-/->/^/v, Alt+<-/-> | Focus neighboring pane

Terminal and console

Key | Function

Ctrl+L | Clear terminal display (when the Terminal tab is active)

Enter | Send command

Ctrl+Shift+M* | Show Compiler Log (problems)

Ctrl+Shift+U* | Show Runtime Output

Build, run and debugging

Key | Function

Ctrl+F5 | Build & Run

Ctrl+Shift+B | Build & Run (identical; no build-only mode)

F5 | "Start debugging": opens the Debugger panel and reports that step-by-step debugging is unavailable

Shift+F5* | Message "Cancelling running programs is not implemented."

F9 | Set/remove breakpoint on the cursor line (Chapter 8)

Panels and windows

Key | Function

Ctrl+B | Toggle side bar

Ctrl+J | Toggle console

Ctrl+Shift+L* | Toggle sidecar

Ctrl+\ , Ctrl+Shift+2* | two columns

Ctrl+Shift+* , Ctrl+Shift+3* | two rows

Ctrl+Shift+4* | 2x2 grid

Ctrl+Shift+W* | close active pane

Ctrl+mouse wheel | UI zoom

Menu shortcuts

The menus show shortcuts only where they appear in the menu text; the entries are listed in Chapter 3. In addition Ctrl+X, Ctrl+C, Ctrl+V and Ctrl+A apply directly in the editor. Help -> Keyboard shortcuts opens a window with the shortcut overview, Help -> NORTEX Code manual opens the manual.

11 First steps

Back · Contents · Next: Efficient workflows

Goal: write a program, build it, fix an error and save.

1. Start: double-click nortex-code.exe. The dashboard appears.
2. Choose the project folder: Activity bar -> Explorer, enter the folder path in the path field (e.g. C:\Nortex\first), Open. The folder must exist; the IDE cannot create a "new project".
3. New file: Activity bar -> Editor, click + in the tab strip. With Ctrl+S or File -> Save as… choose the desired path and file name (extension .nx).
4. Write the program:

```
public int main() {
    int total = 2 + 3;
    return total;
}
```

5. Save: Ctrl+S. Status bar: "File saved".
6. Build and run: Ctrl+F5 (or ? Run). Runtime Output shows "Program finished (exit code Some(5))". The return value of main is the exit code.
7. Provoke an error: change return total; to return totall;. The editor marks the spot; Ctrl+F5 gives NX1004 … Unknown identifier 'totall' in the Compiler Log, status bar red.
8. Fix it: back to total, Ctrl+F5 -> exit code 5 again, status bar green.
9. Add output (optional; one string value per call):

```
using System;

public int main() {
    Console.writeLine("Hello, NORTEX!");
    return 0;
}
```

Runtime Output: Hello, NORTEX!.

10. Save: Ctrl+S. The dot • in the tab disappears as soon as everything is saved.

More runnable examples: examples\language\ (Language reference).

12 Efficient workflows

Back · Contents · Next: FAQ

- Cycle: write -> Ctrl+F5 -> check the Compiler Log -> correct. The IDE saves by itself before every build.
- Command palette (Ctrl+Shift+P) instead of searching menus.
- Window layout: split the editor with Ctrl+\ (source next to example/test). Toggle the console with Ctrl+J, Ctrl+Shift+M/U jumps to the matching tab. Zen mode hides secondary elements.
- Projects: one folder per project, optionally nortex.toml (manifest) for project name, entry file and build options; the window title and status bar then show the project name.
- Navigation: Ctrl+P (open file), project search Ctrl+Shift+F, Ctrl+Tab between tabs, Ctrl+F in the document.
- Edit lines with Alt+^/v, Ctrl+/ and Ctrl+D instead of copy/paste.
- Overview: small functions, meaningful names; auto-completion uses identifiers of the document.
- Always save before closing the IDE; unsaved tabs are not recovered.

13 FAQ and troubleshooting

Back · Contents

Why can't a project be opened? Check the path (the folder must exist). Drag & drop only accepts .nx ("Only .nx files are opened"). A broken nortex.toml produces a message in the status bar.

Why does compilation not work?

1. "Compiler … could not be started": nortexc.exe must be next to nortex-code.exe or on the PATH.
2. Linker/runtime errors: nortex_runtime.lib is missing (next to the EXE, in the parent folder or via NORTEX_RUNTIME_LIB).
3. New, not yet saved files: save with Ctrl+S first and choose a name in the file dialog.
4. NX errors: see Chapter 8.

Where do I find error messages? In the Compiler Log tab (Ctrl+Shift+M), marked in the editor, count in the status bar.

How do I open the terminal? Terminal tab in the bottom panel; show the panel with Ctrl+J.

How do I save files? Ctrl+S or File -> Save; Ctrl+Shift+S for "Save as". Unsaved tabs show •.

How do I change the appearance? Gear -> theme, density, zoom, language (Chapter 9).

Edit several files at once? Tabs and split editor (Chapter 5).

Why does the Run button show "Exitcode Some(0)"? That is how the IDE prints exit code 0; it is not an error.

Why does the IDE report an error for float or struct? These constructs cannot be generated natively yet (NX2001/NX2002).

Is there a debugger, Git commit, rename/delete, AI? No, see the feature status in the contents.

My settings were reset. Delete %APPDATA%\NORTEX Code\settings.json; it is created again.